

Co1 - AT2 - Concept Mapping

Name : Venna Meenakshi Reddy

Reg. No : 192471048

Course
Name : Computer Vision

Course
Code : ITA0501

Slot : A

Concept map

1A)

Overview of Computer Vision

- Enables machines to understand images/video

Uses Cameras & Computer Vision to understand surroundings & make decisions.

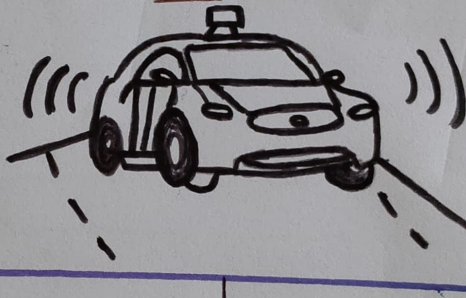
Levels of Computer Vision

- Low-level
- mid-level
- High-level

Features of Computer Vision

- Image acquisition
- Image processing
- Tracking
- Object detection

Autonomous Vehicle



Applications

- Road/Lane detection
- obstacle detection
- Navigations & decision-making

Digital Image Fundamentals

- Pixel
- Resolution
- Intensity
- Grayscale Image
- Colour Image

Image Sequencing & Acquisition

- Camera
- Image sensor
- Lens
- Digital image capture

Sampling & Quantization

- sampling: Converts spatial information into pixels
- quantization: Converts intensity into digital values

Image Formation Models

- Light
- object
- Camera/Lens
- Image formation

Concept Map

2)A)

Overview

- Non-invasive imaging technique.
- Uses x-rays and digital sensors.
- Helps in diagnosis & treatment planning

Uses digital x-ray images to help doctors identify abnormalities in Patients

Medical Imaging

Key features

- Uses x-ray radiation
- Digital Image capture & storage.
- Fast & Accurate.
- Helps detect abnormalities early

Imaging system Capture

- X-ray Source
- Patient (object)
- detector (digital sensor)
- Image acquisition

Digitizes the image

- Converts x-ray signal to digital form
- Sampling
- Quantization

Image Processing

- Noise reduction
- Image enhancement
- Contrast Adjustment
- Edge detection

Feature extraction

- Edges
- Textures
- Patterns
- regions (lungs, nodules etc.)

Object Re-cognition

- Detects objects
- Uses trained models & algorithms.

Digital Image Fundamentals

- Pixel → smallest unit
- Resolution → detail level
- Intensity → brightness value

Applications

- Detecting tumors
- Identifying fractures
- Treatment planning